EVIDENCE LEVEL: E0 STATUS: CONCEPTUAL_PREVIEW MODULE: 6

Travel is a Stress Test

Travel does not make collective market formation easy. It makes it

GOLIAF is a bounded travel stress-test environment. It is not a live booking

1. Why Choose the Hardest Possible Domain?

When evaluating novel market formation mechanisms, the common instinct is to start

However, proving a mechanism on trivial commodities reveals almost nothing about

To understand where collective market positions break down, DC2SC uses the travel

2. The Four Stress Vectors of Travel

In the **GOLIAF** research environment, travel exposes coordination mechanisms to

Non-Averagable Constraint Variance: Household travel constraints (dates,

2. **Temporal Perishability:** Travel inventory perishes as departure dates

3. **Multi-Leg Cascade Risk:** Travel requires coordinating flights,

4. **Catastrophic Dissolution Burden:** Position failure results in stranded

3. Failure as a Primary Research Finding

A foundational principle of DC2SC is that **a stress-test environment is designed

to produce failures**.

In GOLIAF, observing a collective travel position shatter against the reality of

independent date and budget variance.

GOLIAF operates strictly as a bounded stress-test environment, not a commercial booking service.

4. Failure Modes Tracked

Under the **DC2SC Failure-First Framework**, Module 06 focuses on:

Constraint Compression Failure (FAIL-08): Mechanism forcing participants to

Perishability Squeeze (FAIL-09): Delaying collective commitments until

Cascade Disconnection (FAIL-10): Flight coordination succeeding while hotel

5. Research Handoff to Module 07

Understanding that travel stress-testing systematically generates failure brings

It does not constitute commercial claims, supplier commitments, or live transaction guarantees.

MATHEMATICAL INVARIANT // THEOREM 6.1: TRAVEL CONSTRAINT FEASIBILITY FRONTIER (Eq. 6.1)

$$P(\bigcap_{j=1}^{M} C_j(t) \neq q \emptyset) \leq \exp(- M * \beta * Var(T_{dates}) * Var(B_{budget}))$$

Proves why multi-party travel aggregation faces exponential feasibility collapse under

[FORMALLY VERIFIED MATHEMATICAL SPECIFICATION PEER CITATION READY]

CONSTITUTIONAL RESEARCH BOUNDARY & NON-CLAIM DISCLOSURE

This publication presents conceptual research and formal invariants at Evidence Level E0.